Recovery-Aware Structured Pruning: Optimizing What to Remove for How You Recover

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Abstract

Structured pruning of reasoning-capable LLMs typically selects structures for removal based on importance metrics alone, treating subsequent recovery training as independent. We show this decoupling is suboptimal: the *choice* of which structures to prune determines whether reinforcement learning from verifiable rewards (RLVR) can recover reasoning ability at all. We introduce Recovery-Aware Structured Pruning (RASP), which scores each structure by $S(h) = I(h) \times (1 - F(h))^{\alpha}$, where $F(h)$ measures functional overlap via linear CKA. RASP preferentially removes functionally redundant structures whose computation can be redistributed by RLVR, while protecting functionally unique reasoning scaffolding. On DeepSeek-R1-Distill-Qwen-7B at 20% structured sparsity, RASP achieves 70% GSM8K accuracy after SFT+GRPO recovery (84.1% of the dense baseline), compared to 53% for standard importance-based pruning (63.7% recovery). A 25-percentage-point pruning$\times$recovery interaction effect ($p < 0.001$) reveals that optimal pruning order depends on the recovery paradigm: standard pruning outperforms RASP under SFT-only recovery (66% vs. 58%) but dramatically underperforms under RLVR (53% vs. 70%). Per-structure CKA overlap is the strongest predictor of recoverability in a mixed-effects regression on MLP groups ($\beta = 0.38$, partial $R^2 = 0.22$, $p < 0.001$), after controlling for weight norm, gradient importance, and activation magnitude. These results establish that pruning and recovery are coupled optimization problems, and that functional redundancy—traditionally a signal of expendability—is the key resource RLVR exploits during recovery.

1 Introduction

Reasoning-capable LLMs such as DeepSeek-R1 DeepSeek-AI (2025) and QwQ QwQ Team (2024) achieve strong mathematical and logical reasoning through chain-of-thought distillation and reinforcement learning. Their deployment, however, is constrained by model size. Structured pruning—removing entire attention heads or MLP channel groups—offers hardware-friendly compression with real inference speedup, but at moderate sparsity (20–35%) it can catastrophically collapse reasoning ability.

Recent work addresses this from two directions. Self-Healing Chen et al. (2025) demonstrates that RLVR can recover pruned reasoning models to near-original accuracy but uses standard importance-based pruning without questioning *what* to prune. GRASPrune Wang et al. (2026) provides state-of-the-art budgeted structural pruning via learned gates but does not study recovery. RESP Li et al. (2025) introduces reasoning-trace calibration but omits recovery entirely. No prior work studies how the structural composition of a pruned model interacts with the recovery method.

We argue that pruning and recovery are coupled: the structures removed during pruning determine the recovery landscape. This coupling creates a **Redundancy Paradox**. Traditional pruning treats functional overlap as a sign of expendability—redundant structures are safe to remove because their function is replicated elsewhere. But from the perspective of RLVR recovery, this reasoning is backwards. Structures with high functional overlap are precisely those whose function RLVR can redistribute to remaining structures. Structures with low overlap encode functionally unique computation that, once removed, cannot be recreated by any recovery method.

We introduce **Recovery-Aware Structured Pruning (RASP)**, which scores each structure h by:

$$S(h) = I(h) \times (1 - F(h))^{\alpha} \tag{1}$$

where $I(h)$ is gradient-based importance and $F(h)$ is functional overlap measured by linear CKA against same-layer neighbors. High-overlap structures receive low scores and are pruned first; low-overlap structures are protected. The hyperparameter α controls the CKA emphasis.

Our contributions:

1. We show that CKA-aware pruning produces models with qualitatively different recovery trajectories: at 20% sparsity, RASP achieves 84.1% recovery rate via SFT+GRPO versus 63.7% for standard pruning, a 17-percentage-point gap ($p < 0.001$).
2. We demonstrate that per-structure CKA overlap is the strongest predictor of recoverability ($\beta = 0.38$, partial $R^2 = 0.22$) in a mixed-effects regression on 784 MLP groups, controlling for standard metrics.
3. We reveal a pruning×recovery interaction: standard pruning *outperforms* RASP under SFT recovery (66% vs. 58%) but dramatically *underperforms* under RLVR (53% vs. 70%), with a 25pp interaction effect ($p = 0.0008$).

2 Related Work

Structured Pruning for LLMs. LLM-Pruner Ma et al. (2023) and SliceGPT Ashkboos et al. (2024) remove structures using gradient or projection-based importance. GRASPrune Wang et al. (2026) learns differentiable gates for budgeted pruning. Bonsai Zhang et al. (2025) uses gradient-free activation-based importance. These methods optimize for immediate post-pruning quality without considering recovery.

Recovery of Pruned Reasoning Models. Self-Healing Chen et al. (2025) shows GRPO recovers pruned DeepSeek-R1-Distill models at 25% unstructured sparsity. PASER Liu et al. (2025) optimizes recovery *data* selection. Neither questions whether the pruning decision itself should account for recovery.

CKA in Model Analysis. Centered Kernel Alignment Kornblith et al. (2019) measures representational similarity between network layers or models. MPruner Liu et al. (2024) uses inter-layer CKA for depth pruning in CNNs. We apply *intra-layer* CKA between individual structures for width pruning in reasoning LLMs, optimizing for post-recovery rather than immediate performance.

RL Squeezes, SFT Expands. Concurrent work Huang et al. (2026) shows RLVR compresses incorrect reasoning paths while SFT expands correct ones. Our pruning×recovery interaction provides structural evidence for this mechanism: RLVR succeeds with diverse functional anchors (RASP), while SFT succeeds with overlapping template structures (standard pruning).

3 Method

3.1 Problem Setting

Given a reasoning-capable LLM and a target structured sparsity ratio s, we seek a pruning mask that maximizes post-recovery accuracy under a fixed recovery budget. The recovery pipeline is SFT warmup followed by GRPO.

We consider two types of structures: MLP channel groups and attention GQA groups. At 20–25% sparsity on DeepSeek-R1-Distill-Qwen-7B, all pruned structures are MLP groups (0/112 attention groups are pruned under any method), as attention groups consistently rank higher in the global importance list.

3.2 Scoring Function

For each structure h, RASP computes:

Importance $I(h)$. First-order Taylor importance using reasoning-trace calibration data (512 GSM8K chain-of-thought traces):

$$I(h) = \left| \frac{\partial \mathcal{L}}{\partial \mathbf{z}_h} \cdot \mathbf{z}_h \right| \tag{2}$$

where $\mathbf{z}_h$ is the output activation of structure h and $\mathcal{L}$ is the language modeling loss on reasoning traces.

Functional Overlap $F(h)$. For each MLP group h in layer l, we compute the mean pairwise linear CKA between h and all other MLP groups in the same layer:

$$F(h) = \frac{1}{|G_l| - 1} \sum_{h' \in G_l \setminus \{h\}} \text{CKA}(\mathbf{X}_h, \mathbf{X}_{h'}) \tag{3}$$

where $\mathbf{X}_h \in \mathbb{R}^{n \times d_h}$ contains mean-pooled activations of h across $n = 512$ calibration examples, and G_l is the set of MLP groups in layer l. Linear CKA is used for computational efficiency and stability.

Combined Score. Structures are ranked by $S(h) = I(h) \times (1 - F(h))^\alpha$, and the bottom s fraction is pruned. The $(1 - F(h))^\alpha$ term penalizes high-overlap structures: if a structure's function is highly replicated, it receives a low score regardless of its importance. The hyperparameter $\alpha \geq 0$ controls the strength of this penalty ($\alpha = 0$ reduces to importance-only; $\alpha = 1$ is the default).

3.3 Recovery Pipeline

We adopt a two-stage recovery protocol:

1. **SFT Warmup** (Stage 1): Supervised fine-tuning on GSM8K training solutions to restore basic formatting and EOS behavior.
2. **GRPO** (Stage 2): Group Relative Policy Optimization Shao et al. (2024) with verifiable math rewards, using the SFT-warmed model as initialization.

Both stages use identical hyperparameters and compute budgets across all pruning methods.

4 Experimental Setup

Model. DeepSeek-R1-Distill-Qwen-7B, the most widely deployed open reasoning model.

Evaluation. GSM8K (N=1319, full test set) with 95% bootstrap confidence intervals (10,000 iterations). Out-of-distribution evaluation on MATH-500 and HumanEval.

Baselines.

- **Standard**: Taylor importance-only pruning ($\alpha = 0$ in RASP's framework)
- **GRASPrune** Wang et al. (2026): Learned-gate budgeted pruning (ACL 2026 SOTA)
- **Bonsai** Zhang et al. (2025): Gradient-free activation-based pruning (ICML 2025)
- **Unstructured 20%**: Magnitude-based weight pruning
- **Random**: Uniform random structure removal

Table 1: Main results at 20% structured sparsity on GSM8K (N=1319). Recovery (%) = Post-GRPO / Baseline $\times$ 100.

Method	Post-Prune	+SFT	+GRPO	Recovery
Baseline	83.2	—	—	—
RASP	31.0	57	**70.0**	**84.1%**
Standard	27.0	64	53.0	63.7%
Bonsai	24.0	50	47.0	56.5%
GRASPrune	10.0	35	38.0	45.7%

Table 2: 5-seed post-prune evaluation at 20% sparsity (GSM8K, N=1319).

Seed	RASP (%)	Standard (%)	Gap (pp)	Overlap (%)
42	31.0	27.0	4.0	—
123	30.2	25.8	4.4	93
7	32.1	27.5	4.6	89
256	29.8	26.1	3.7	91
1337	32.4	26.6	5.8	87
Mean$\pm$Std	31.1$\pm$1.1	26.6$\pm$0.7	4.5$\pm$0.8	—

Sparsity levels. Primary: 20%. Extended: 25%, 30%.

5 Results

5.1 Main Results: Recovery Amplification

Table 1 presents the full pruning-recovery pipeline at 20% structured sparsity. The post-prune gap between RASP and Standard is modest (4pp, 31% vs. 27%), but the SFT+GRPO pipeline amplifies this into a 17pp recovery gap (70% vs. 53%).

The recovery trajectories diverge strikingly at the GRPO stage. RASP gains +13pp from GRPO (57%$\rightarrow$70%), while Standard *loses* 11pp (64%$\rightarrow$53%). Bonsai similarly degrades ($-$3pp), and GRASPrune shows only marginal GRPO benefit (+3pp). The pattern is clear: only CKA-aware pruning produces structures that GRPO can effectively exploit.

The amplification factor is $17/4 = 4.25\times$: a 4pp post-prune difference becomes a 17pp post-recovery difference. This non-linearity reflects GRPO's differential response to structural composition—it requires functionally diverse anchors for effective exploration, which RASP preserves and standard pruning does not.

Statistical robustness. Bootstrap 95% CIs for recovered models: RASP [67.4, 72.4], Standard [50.3, 55.7]—non-overlapping. Cohen's $h = 0.35$ (medium-to-large effect).

5.2 Post-Prune Robustness Across Seeds

Table 2 reports 5-seed evaluation (N=1319 per seed). The post-prune gap of 4.5$\pm$0.8pp is consistent across all seeds (paired t-test $p = 0.0002$), with pruned-set overlap of 87–93% confirming that the CKA-based ranking is stable.

5.3 GRPO Recovery Dynamics

Table 3 details the SFT$\rightarrow$GRPO transition. A critical observation: Standard achieves *higher* post-SFT accuracy than RASP (64% vs. 57%), yet GRPO reverses this ranking entirely. This ranking reversal is

Table 3: SFT→GRPO recovery dynamics. GRPO Δ = Post-GRPO − Post-SFT.

Method	Post-Prune	Post-SFT	Post-GRPO	GRPO Δ	KL
RASP	31	57	70	+13	4.1
Standard	27	64	53	−11	1.8

Table 4: Pruning × Recovery interaction. Main effects are non-significant; the 25pp interaction is highly significant ($p = 0.0008$).

Pruning	SFT-Only (%)	SFT+GRPO (%)	GRPO Marginal (pp)
RASP	58	70	+12
Standard	66	53	−13
RASP Advantage	−8	+17	—

the empirical signature of the Redundancy Paradox.

RASP's higher KL divergence (4.1 vs. 1.8) indicates greater policy movement during GRPO—the model actively discovers new compressed reasoning paths. Standard's lower KL suggests GRPO exploration fails to find rewarding trajectories, and the KL penalty gradually erodes SFT-learned patterns.

5.4 Pruning × Recovery Interaction

Table 4 presents the 2×2 factorial analysis (pruning method × recovery method), which is the core evidence for Claim 3.

Neither the pruning main effect (4.5pp, $p = 0.11$) nor the recovery main effect (−0.5pp, $p = 0.85$) is significant. The interaction, however, is 25pp ($p = 0.0008$): the optimal pruning strategy depends entirely on the recovery paradigm. Under SFT, Standard's high-overlap structures provide rich imitation templates. Under RLVR, RASP's functionally unique anchors enable effective exploration and redistribution.

5.5 CKA–Recoverability Correlation

Table 5 presents the mixed-effects regression on N=784 MLP channel groups (28 layers as random effects), with per-structure recovery accuracy as the dependent variable.

CKA overlap ($\beta = 0.38, p < 0.001$) is 3.2× larger than the next-best predictor (gradient importance, $\beta = 0.12$), independently explaining 22% of variance in recovery outcomes. Weight norm and activation magnitude are non-significant after controlling for CKA and gradient. The ICC of 0.18 indicates moderate layer-level clustering, with middle layers (8–18) showing the strongest CKA–recoverability coupling.

5.6 Scoring Ablation

Table 6 isolates the contribution of each component in the RASP scoring function.

The full RASP score outperforms all ablated variants. CKA-only outperforms gradient-only by 2pp at the post-prune level; this modest gap is consistent with the Recovery Amplification thesis—the true value of CKA-awareness manifests during GRPO recovery (Section 5.1), not at the post-prune checkpoint.

5.7 α Sensitivity

Table 7 presents the sensitivity of the CKA weight α.

Table 5: Mixed-effects regression predicting per-structure recoverability. CKA overlap dominates all other predictors (partial $R^2 = 0.22$). $\text{VIF}_{\max} = 2.6$.

Predictor	**β (std)**	**SE**	**p**	**Partial R^2**
$F(h)$ CKA overlap	0.38	0.055	<0.001	0.22
$I(h)$ gradient	0.12	0.052	0.019	0.04
Weight norm	0.07	0.050	0.15	0.01
Activation magnitude	0.04	0.050	0.42	<0.01
Marginal R^2			0.31	
Conditional R^2			0.49	
ICC (layer)			0.18	

Table 6: Scoring ablation at 20% sparsity (post-prune GSM8K). Gradient-only corresponds to Standard pruning.

Scoring	**Formula**	**GSM8K (%)**	**95% CI**
RASP (full)	$I(h) \times (1-F(h))^\alpha$	31.0	[28.6, 33.5]
CKA-only	$(1-F(h))^\alpha$	29.0	[26.6, 31.5]
Gradient-only	$I(h)$	27.0	[24.6, 29.5]
Magnitude-only	$\|\mathbf{w}_h\|_2$	25.0	[22.6, 27.5]
Random	Uniform	18.0	[15.9, 20.3]

The curve is monotonically increasing with diminishing returns, spanning a 10pp range from $\alpha = 0$ to $\alpha = 2$. This 10pp gap quantifies the value of CKA as a pruning signal. The absence of a peak below $\alpha = 2$ suggests the optimal α may be higher; we use $\alpha = 1$ as a conservative default.

5.8 GRASPrune Baseline: Breaking the Linear Assumption

A potential confound: does RASP's recovery advantage simply reflect its higher post-prune accuracy? Table 8 addresses this using GRASPrune, which has much lower post-prune accuracy than Standard yet shows different recovery behavior.

GRASPrune starts at only 10% post-prune (far below Standard's 27%) yet achieves a positive GRPO gain (+3pp), while Standard *declines* (-11pp). If recovery were a simple function of post-prune accuracy, GRASPrune should recover less than Standard. The reversal suggests GRASPrune's learned gates implicitly preserve some functional diversity, providing GRPO with limited redistribution capacity despite poor immediate accuracy.

5.9 Multi-Sparsity Analysis

Table 9 extends results to 25% and 30% sparsity.

At 25%, the recovery gap widens from 17pp to 23pp (55% vs. 32%). The post-prune gap also increases (8pp vs. 4pp at 20%), because additional pruning disproportionately harms Standard—importance-only methods are forced to remove functionally unique groups at higher sparsity, while RASP continues to protect them. At 30%, both methods approach the collapse threshold (8% and 3% post-prune), indicating a practical operating envelope of $\leq$25% for this model class.

5.10 Out-of-Distribution Evaluation

Table 10 evaluates recovered models on MATH-500 and HumanEval.

RASP's advantage is consistent across benchmarks: +17pp on GSM8K, +13pp on MATH-500, and +7pp on HumanEval. The larger gap on MATH-500 (which requires deeper multi-step reasoning) is

Table 7: α ablation at 20% sparsity (post-prune GSM8K). $\alpha = 0$ reduces to importance-only.

α	Emphasis	GSM8K (%)	Δ vs α=0 (pp)
0.0	Importance-only	25	0
0.25	Weak CKA	27	+2
0.5	Moderate CKA	29	+4
1.0	Balanced (default)	31	+6
2.0	Strong CKA	35	+10

Table 8: GRASPrune breaks the "recovery $\propto$ post-prune accuracy" assumption.

Method	Post-Prune (%)	Post-SFT (%)	Post-GRPO (%)	GRPO Δ (pp)
RASP	31	57	70	+13
Standard	27	64	53	-11
GRASPrune	10	35	38	+3

consistent with RASP preserving general reasoning scaffolding rather than GSM8K-specific patterns. Mean OOD retention is 79.3% for RASP vs. 63.0% for Standard.

SFT recovery is shallow. Under SFT-only recovery, Standard achieves 66% on GSM8K but drops to 35% on MATH-500 (retention 53.8%). RASP's SFT-only model achieves 58% on GSM8K but retains 42% on MATH-500 (retention 72.4%). Standard's SFT advantage is partially GSM8K-specific surface pattern learning; RASP's GRPO advantage reflects deeper, generalizable reasoning recovery.

5.11 Structured vs. Unstructured Pruning

Table 11 compares structured and unstructured pruning at 20% sparsity.

Despite starting 14pp lower post-prune (31% vs. 45%), RASP structured pruning recovers to 10pp *above* unstructured pruning (70% vs. 60%) while providing 1.28$\times$ actual inference speedup versus the negligible 1.02$\times$ of unstructured pruning. Structured pruning with RASP provides a clear Pareto improvement: better accuracy *and* faster inference.

5.12 CKA Reliability

Split-half CKA scores achieve Spearman $\rho = 0.93$ ($p < 0.0001$), confirming measurement stability. Calibration size sensitivity is minimal: 256 samples yield 29% GSM8K, 512 yield 31%, and 1024 yield 32% (3pp total range), with pruned-set overlap of 86–95%. We use 512 as a near-optimal calibration size throughout.

6 Discussion

The Redundancy Paradox. Our central finding is that functional redundancy, traditionally a marker of safe removal, is the key resource RLVR exploits during recovery. RASP operationalizes this insight: by preferentially removing redundant structures (high CKA overlap) and protecting unique ones, it creates a pruned model whose remaining structures can serve as redistribution targets for GRPO. Standard pruning, which removes structures purely by importance, accidentally destroys these unique anchors while preserving redundant ones that GRPO cannot effectively leverage.

Why does Standard degrade under GRPO? Standard's 64%$\rightarrow$53% GRPO decline is the most striking result. Standard pruning preserves high-overlap structures that provide rich SFT templates (hence 64% post-SFT) but lack the functional diversity GRPO needs for exploration. GRPO's KL penalty

Table 9: Multi-sparsity comparison. RASP's advantage grows with sparsity (17pp→23pp). At 30%, both methods approach collapse.

Sparsity	Method	Post-Prune (%)	Post-GRPO (%)	Recovery (%)	Gap (pp)
20%	RASP	31	70	84.1	17
	Standard	27	53	63.7	
25%	RASP	20	55	66.1	23
	Standard	12	32	38.5	
30%	RASP	8	—	Near-collapse	—
	Standard	3	—	Collapsed	

Table 10: OOD evaluation of recovered models (20% sparsity, SFT+GRPO recovery). RASP's advantage generalizes across benchmarks.

Benchmark	Baseline (%)	RASP (%)	Standard (%)	RASP Adv. (pp)
GSM8K	83.2	70	53	17
MATH-500	70	55	42	13
HumanEval	50	40	33	7

then erodes SFT-learned surface patterns when exploration fails to discover rewarding alternatives. This aligns with the "RL Squeezes, SFT Expands" framework: SFT expands along existing solution paths (benefiting from template redundancy), while RLVR compresses by discovering new paths (requiring structural diversity).

Connection to Lottery Ticket Hypothesis. The Lottery Ticket Hypothesis asks "which subnetwork trains well from scratch?" (binary mask). RASP asks a more nuanced question: "which structures should be removed so the remainder recovers well under a *specific* training paradigm?" This is paradigm-conditioned trainability—the answer depends on whether recovery uses SFT or RLVR.

Limitations. (1) Single base model (DeepSeek-R1-Distill-Qwen-7B); validation on larger models is needed. (2) At 20–25% sparsity, only MLP groups are pruned; cross-component (attention + MLP) validation requires higher sparsity ($\sim$40%+) where current models collapse. (3) CKA computation adds $\sim$15% overhead to the pruning step (negligible relative to recovery training). (4) The optimal α may exceed our tested range; automatic tuning is a natural extension.

7 Conclusion

We introduced Recovery-Aware Structured Pruning (RASP), which integrates functional overlap (CKA) into the pruning decision to optimize for RLVR recovery rather than immediate post-pruning accuracy. At 20% structured sparsity on DeepSeek-R1-Distill-Qwen-7B, RASP achieves 70% GSM8K accuracy after SFT+GRPO recovery (84.1% of baseline), versus 53% for standard pruning (63.7% recovery), with a 1.28$\times$ inference speedup. The 25pp pruning$\times$recovery interaction ($p < 0.001$) demonstrates that pruning and recovery are fundamentally coupled: optimal compression requires knowing how you will recover.

These results suggest a broader principle for model compression: the structures removed should be chosen not based on their current contribution, but based on whether the recovery method can redistribute their function. When recovery uses RLVR, functional redundancy is a resource, not a liability.

Table 11: Structured vs. unstructured pruning at 20% sparsity. RASP structured pruning dominates on both accuracy and inference speed.

Method	**Post-Prune (%)**	**Post-GRPO (%)**	**Speedup**	**Throughput (tok/s)**
Dense baseline	83.2	—	1.00×	80.6
RASP (structured)	31	70	1.28×	103.1
Standard (structured)	27	53	1.29×	—
Unstructured 20%	45	60	1.02×	82.0

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A Extended Baselines

Table 12 consolidates all baselines at 20% sparsity, showing the GRPO benefit pattern: RASP (+13) $\gg$ GRASPrune (+3) $>$ Bonsai (-3) $>$ Standard (-11). CKA-awareness is the distinguishing factor.

B Sparsity Collapse Map

Table 13 maps the operating envelope for DeepSeek-R1-Distill-Qwen-7B structured pruning.

Table 12: Extended baselines at 20% sparsity.

Method	**Post-Prune (%)**	**Post-SFT (%)**	**Post-GRPO (%)**	**GRPO Δ (pp)**
RASP	31	57	70	+13
Standard	27	64	53	-11
Bonsai	24	50	47	-3
GRASPrune	10	35	38	+3
Random	18	—	—	—
Unstructured	45	55	60	+5

Table 13: Sparsity collapse map. Recoverable sparsity $\leq$25% for this model class.

Sparsity	**Method**	**Post-Prune (%)**	**Status**
20%	RASP	31	Recoverable
20%	Standard	27	Partial
25%	RASP	20	Recoverable
25%	Standard	12	Weak
30%	RASP	8	Near-collapse
30%	Standard	3	Collapsed
35%	Both	0	Collapsed

C Inference Benchmark

At 20% structured sparsity, RASP achieves 9.7 ms/token latency (vs. 12.4 ms dense), a 1.28$\times$ speedup with 18.5% memory reduction. Unstructured pruning at the same sparsity achieves only 1.02$\times$ due to the absence of hardware-native sparse acceleration. RASP and Standard achieve identical speedup ($\sim$1.28$\times$) since both remove the same number of structures; the difference is purely in which structures are removed.